The Fourier Slice Theorem: Statement, Proof, Intuition, and Visualizations

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Abstract

This note presents the Fourier Slice Theorem (also called the Central Slice Theorem), gives a formal proof under standard integrability assumptions, explains the geometric intuition through visualizations, and connects the result to image reconstruction in computed tomography (CT). We use the formulation

$$\underbrace{\mathcal{F}_t\{p_\theta(t)\}(\omega)}_{\text{LHS: Fourier transform of projection}} = \underbrace{F(\omega\cos\theta,\, \omega\sin\theta)}_{\text{RHS: slice of 2D Fourier transform of object}}.$$

1 Setup and Notation

Let $f(x, y)$ be a 2D object (image intensity / attenuation map), with $f \in L^1(\mathbb{R}^2)$. Define its 2D Fourier transform as

$$F(k_x, k_y) = \iint_{\mathbb{R}^2} f(x, y)\, e^{-i2\pi(k_x x + k_y y)}\, dx\, dy.$$

For angle θ, define unit vectors

$$\boldsymbol{u}_\theta = (\cos\theta, \sin\theta), \qquad \boldsymbol{v}_\theta = (-\sin\theta, \cos\theta),$$

so any point can be written as

$$(x, y) = t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta.$$

The projection (Radon transform at angle θ) is

$$p_\theta(t) = \int_{-\infty}^{\infty} f\big(t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta\big)\, ds.$$

2 Fourier Slice Theorem

Theorem 1 (Fourier Slice Theorem)**.** *Let $f \in L^1(\mathbb{R}^2)$ and let $p_\theta(t)$ be its projection at angle θ. Then the 1D Fourier transform of p_θ with respect to t equals the restriction of F to the radial line through the origin at angle θ:*

$$\mathcal{F}_t\{p_\theta(t)\}(\omega) = \int_{-\infty}^{\infty} p_\theta(t) e^{-i2\pi\omega t}\, dt = F(\omega\cos\theta, \omega\sin\theta).$$

Higher-dimensional operator form. The Fourier Slice Theorem extends naturally to higher dimensions. In operator notation, it may be written as

$$F_m P_m = S_m F_n,$$

where P_m denotes the projection operator, F_n and F_m denote the n- and m-dimensional Fourier transforms respectively, and S_m denotes restriction (slicing) of the n-dimensional Fourier transform to the appropriate m-dimensional subspace. In words: the Fourier transform of a projection equals a slice of the Fourier transform of the original object.

3 Formal Proof

Start from the definition of $p_\theta(t)$:

$$p_\theta(t) = \int_{-\infty}^{\infty} f\big(t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta\big)\,ds.$$

Take the 1D Fourier transform in t:

$$\mathcal{F}_t\{p_\theta\}(\omega) = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f\big(t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta\big)\,ds \right) e^{-i2\pi\omega t}\,dt.$$

By Fubini's theorem (valid under L^1 assumptions), interchange integrals:

$$\mathcal{F}_t\{p_\theta\}(\omega) = \iint_{\mathbb{R}^2} f\big(t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta\big) e^{-i2\pi\omega t}\,ds\,dt.$$

Use rotated coordinates

$$\begin{pmatrix} x \\ y \end{pmatrix} = t\,\boldsymbol{u}_\theta + s\,\boldsymbol{v}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} t \\ s \end{pmatrix},$$

whose Jacobian determinant is 1, so $dx\,dy = dt\,ds$. Also,

$$t = x\cos\theta + y\sin\theta.$$

Therefore,

$$\begin{aligned} \mathcal{F}_t\{p_\theta\}(\omega) &= \iint_{\mathbb{R}^2} f(x,y) \exp\left(-i2\pi\omega(x\cos\theta + y\sin\theta)\right)\,dx\,dy \\ &= \iint_{\mathbb{R}^2} f(x,y) \exp\big(-i2\pi\big((\omega\cos\theta)x + (\omega\sin\theta)y\big)\big)\,dx\,dy \\ &= F(\omega\cos\theta, \omega\sin\theta). \end{aligned}$$

This proves the theorem.

4 Intuition

The theorem says: if you sum the image along parallel lines at angle θ (a projection), then Fourier-transform that 1D signal, you get exactly one radial line (slice) through the 2D Fourier transform of the image at the same angle.

Equivalent interpretation:

- LHS: compress in space first (projection), then transform in 1D.
- RHS: transform in 2D first, then sample along a line through the origin.

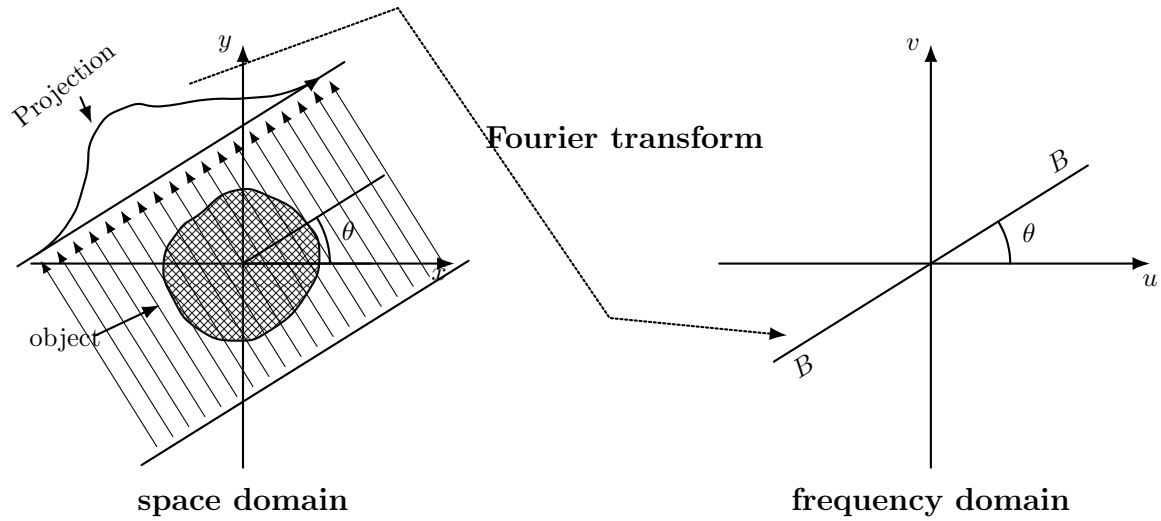


Figure 1: Geometric interpretation of the Fourier Slice Theorem. A parallel projection of $f(x, y)$ taken at angle θ produces a one-dimensional signal $p_\theta(t)$ along the projection axis. The Fourier transform of this signal equals the values of the two-dimensional Fourier transform $F(u, v)$ along a radial line through the origin at the same angle θ. Adapted from Kak and Slaney [1].

5 Connection to LHS/RHS Formulation

We use the notation LHS/RHS for the left and right hand sides respectively,

$$\mathrm{LHS}(\omega, \theta) = \log\big(1 + |\mathcal{F}_t\{p_\theta\}(\omega)|\big),$$

$$\mathrm{RHS}(\omega, \theta) = \log\big(1 + |F(\omega\cos\theta, \omega\sin\theta)|\big).$$

The theorem is exact before magnitude/log scaling; in practice, visual comparisons are often shown with $\log(1 + |\cdot|)$ for dynamic-range compression.

Remark 1. *In discrete numerical implementations (e.g., rotated-grid interpolation, finite sampling, FFT grid mismatch), small discrepancies between LHS and RHS are expected, even though the continuous theorem is exact.*

6 Application: Computed Tomography (CT)

In X-ray CT, each detector reading approximates a line integral of attenuation through the body. For each view angle θ, the scanner acquires a projection $p_\theta(t)$. By the Fourier Slice Theorem:

1. Take 1D Fourier transforms of measured projections.
2. Place each transformed projection as a radial line in 2D frequency space.
3. After collecting many angles, recover $F(k_x, k_y)$ over the plane.
4. Invert the 2D Fourier transform (or use filtered backprojection[1]) to reconstruct $f(x, y)$.

[1]Filtered backprojection (FBP) is the standard reconstruction algorithm in CT. See Kak and Slaney, *Principles of Computerized Tomographic Imaging*,

Because real scanners operate with finite detector spacing, finite angular sampling, and noisy measurements, reconstruction is only an approximation of the continuous object. The quality of the reconstruction therefore depends strongly on detector resolution, angular coverage, and sampling density.

7 Visualizations

A live visualization is available at: `https://mullachv-fourier-slice-app-ba3mcs.streamlit.app/`

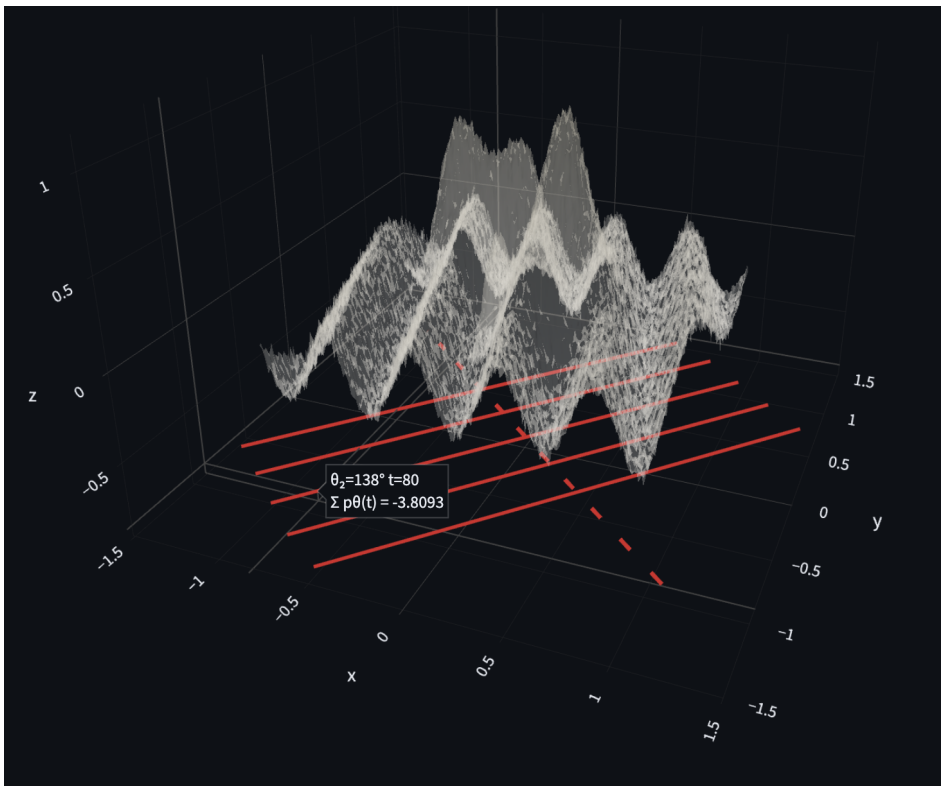


Figure 2: Spatial domain: f(x,y) as a 3D surface + projection lines (θ_1 and θ_2)

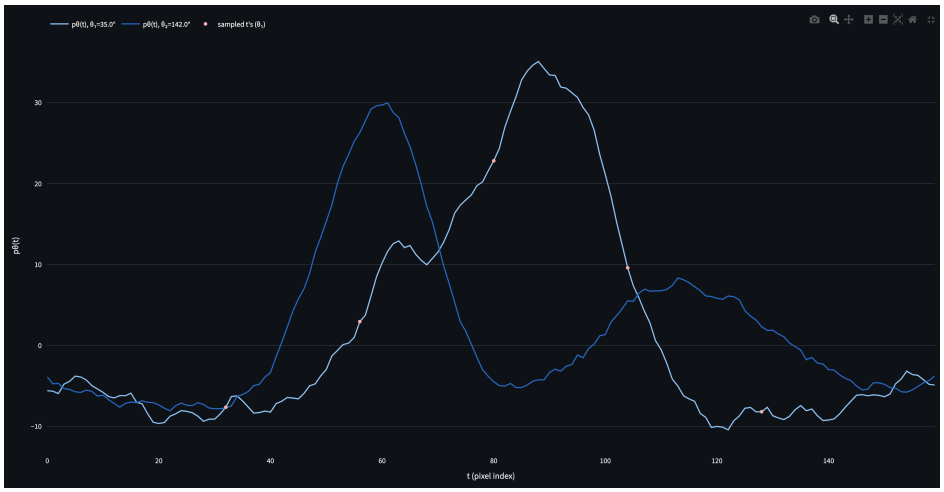
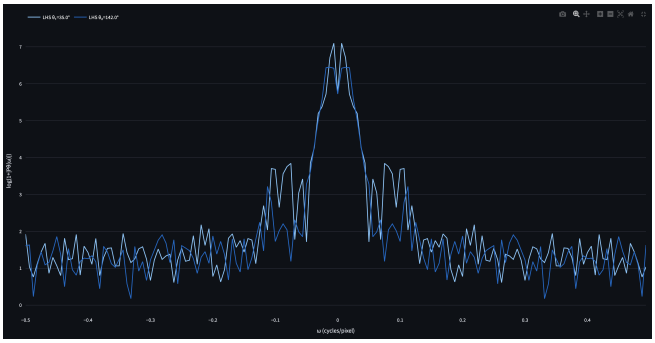
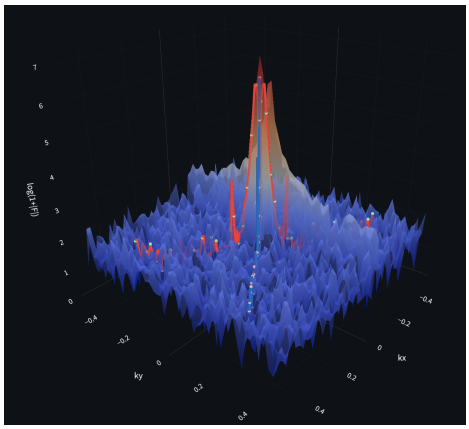


Figure 3: Projections: $p_\theta(t)$ for θ_1 and θ_2



(a) LHS curves $\log(1 + |\mathcal{F}\{p_\theta\}|)$ for θ_1 and θ_2



(b) RHS slices on $\log(1+|F(k_x, k_y)|)$ surface

Figure 4: Comparison of Fourier-transform-of-projection (LHS) and slice-of-2D-Fourier (RHS).

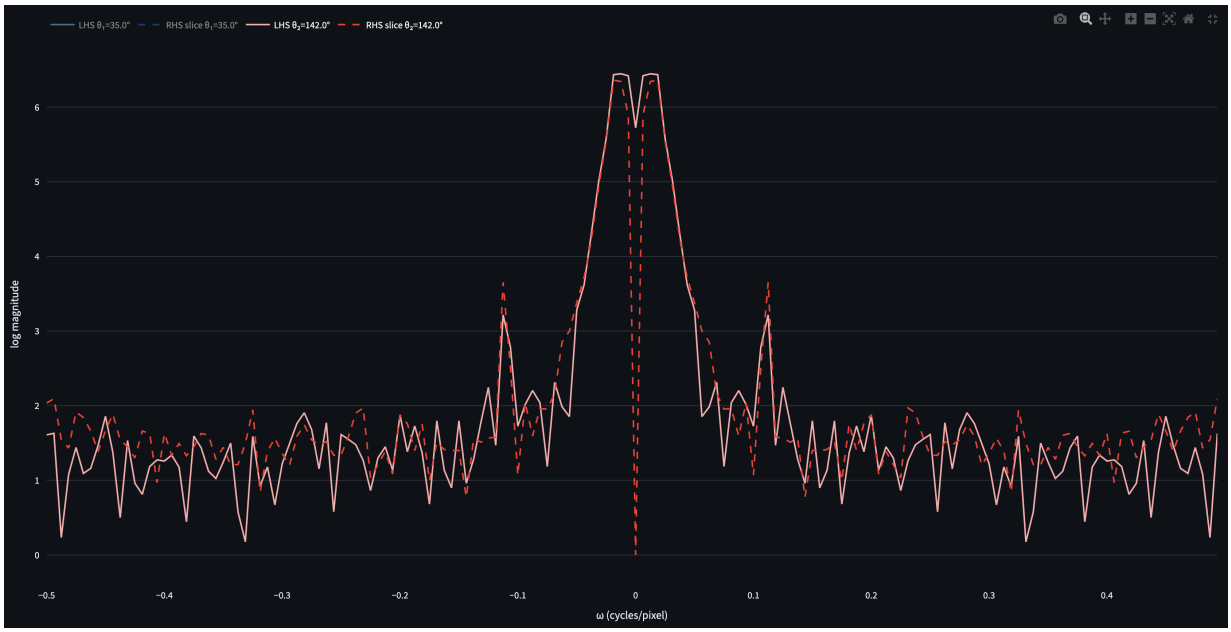


Figure 5: Overlay plot of LHS vs RHS along radial ω

8 Conclusion

The Fourier Slice Theorem provides the mathematical bridge between projection data and frequency-domain image content. It is both a theoretical identity and a practical algorithmic foundation for tomographic imaging, especially CT.

References

1. A. C. Kak and M. Slaney, *Principles of Computerized Tomographic Imaging*, SIAM, 2001.
2. F. Natterer, *The Mathematics of Computerized Tomography*, SIAM, 2001.
3. Wikipedia contributors, *Projection-slice theorem*, `https://en.wikipedia.org/wiki/Projection-slice_theorem`.